

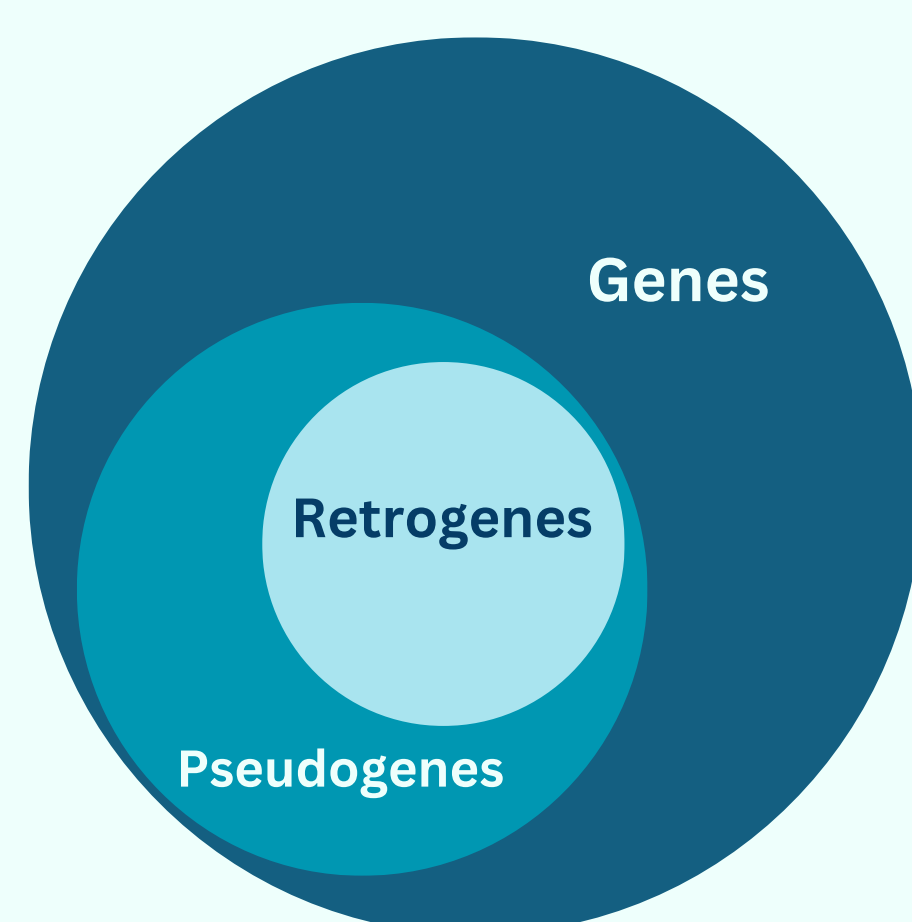
# PCNA Gene Copy Number Variation in Cetaceans

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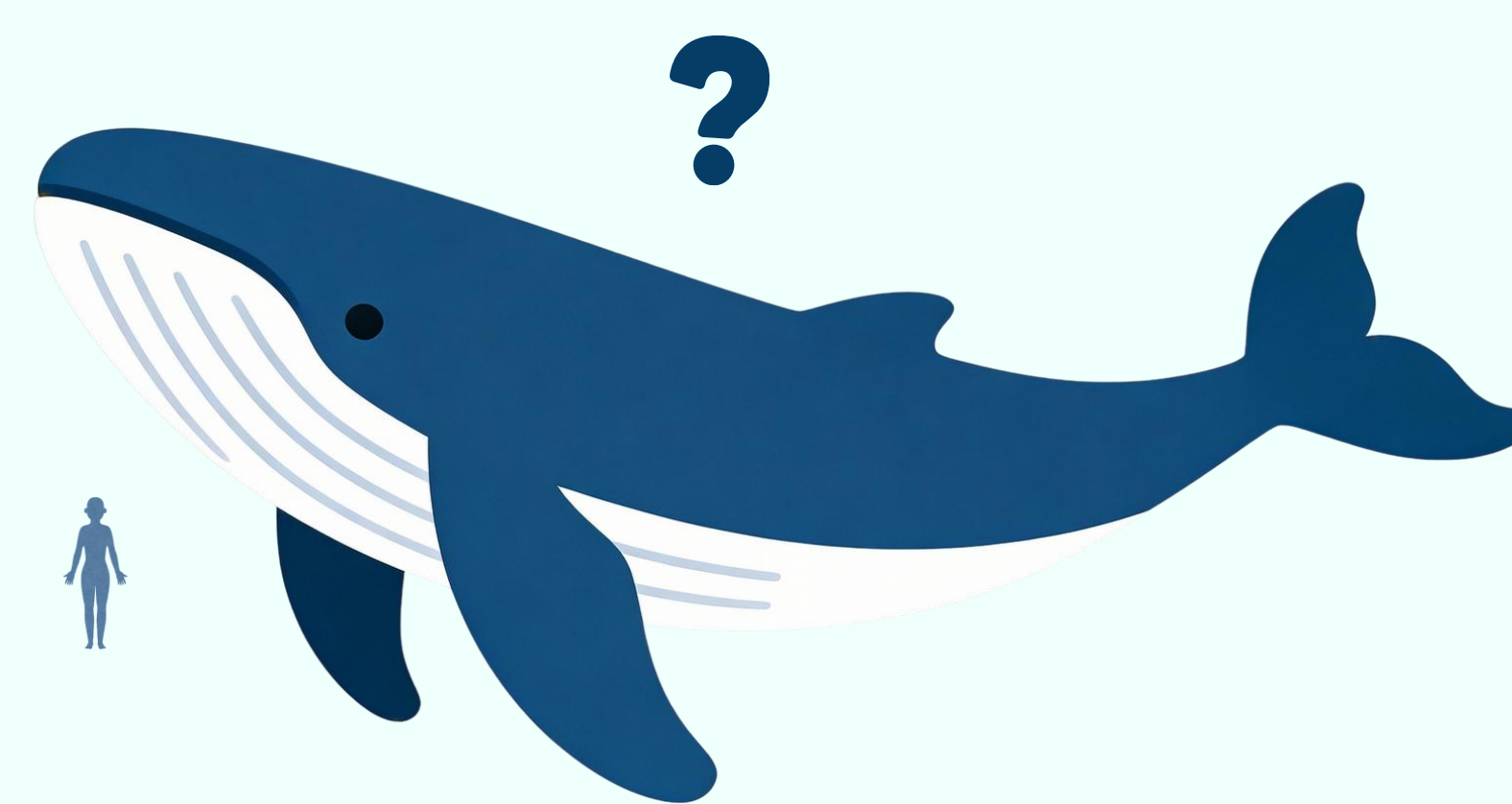
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## BACKGROUND

- **Peto's paradox:** The rate of cancer varies between species in an unexpected manner. More cells and longer lifetime should mean more cancer, but this does not seem to be the case in nature.<sup>2,4</sup>
- One prevalent hypothesis is **copy number variation of cancer-resistance genes**, particularly through **retrotransposition**.<sup>3</sup>
- The **Proliferating Cell Nuclear Antigen (PCNA)** gene duplication has been previously reported in the *Balaena mysticetus* (bowhead whale), one of the largest and longest living mammals.<sup>1</sup>



When gene copies are formed through retrotransposition, they characteristically lack introns and are termed retrogenes. These are mostly non-functional. However, some may exhibit novel functions.<sup>5</sup>



## GOAL

Investigate PCNA duplications in cetaceans to better understand evolutionary patterns and cancer resistance

## METHODS

Compared 2 sub-groups in Cetacea:  
**Odontoceti (toothed whales) vs. Mysticeti (baleen whales)**

1. *Delphinapterus leucas* (Beluga whale)
2. *Eubalaena glacialis* (North Atlantic right Whale)
3. *Tursiops truncatus* (Bottlenose dolphin)
4. *Balaenoptera musculus* (Blue whale)

**Purpose of this study:**  
Validation of existing data and future methods

1. Assess existing publicly available data: NCBI = primary resource

NCBI BLASTn, tBLASTn

2. Long range PCR on samples to select and amplify specific copies

Evaluation of sequences obtained from different primer designs

3. Amplicon sequencing on the PCR products: Oxford Nanopore long-reads

Nextflow pipeline w/ amplicon. 3 modes of usage: de-novo, variant calling and iterative polishing

## RESULTS

1. Toothed whales seem to have 2 full retrogene copies, whereas baleen whales only have 1

2. Primers designed 150 bp outside area-of-interest were most successful in selecting whole copy sequence

3. All pipeline modes contribute to generation and confirmation of consensus sequence

| Species            | Group   | Copy evidence  | % Identity to parent PCNA |
|--------------------|---------|----------------|---------------------------|
| NA right whale     | Baleen  | retrogene (RG) | 97.7%                     |
| Beluga whale       | Toothed | RG1, RG2       | 99.0%, 96.4%              |
| Blue whale         | Baleen  | RG             | 98.0%                     |
| Bottlenose dolphin | Toothed | RG1, RG2       | 98.5%, 93.4%              |

Table 1: Overall comparison of species sampled and their similarity to their parent PCNA exonic sequence

| position  | 26         | 34         | 38         | 58         | 146        | 192        | 244        |
|-----------|------------|------------|------------|------------|------------|------------|------------|
| predicted | Ala to Asp | Gly to Val | Gln to His | Asp to Ser | Arg to Cys | Glu to Val | Met to Thr |

Table 2: The predicted protein from the NA right whale PCNA retrogene shows 18 nucleotide substitutions resulting in 7 amino-acid substitutions

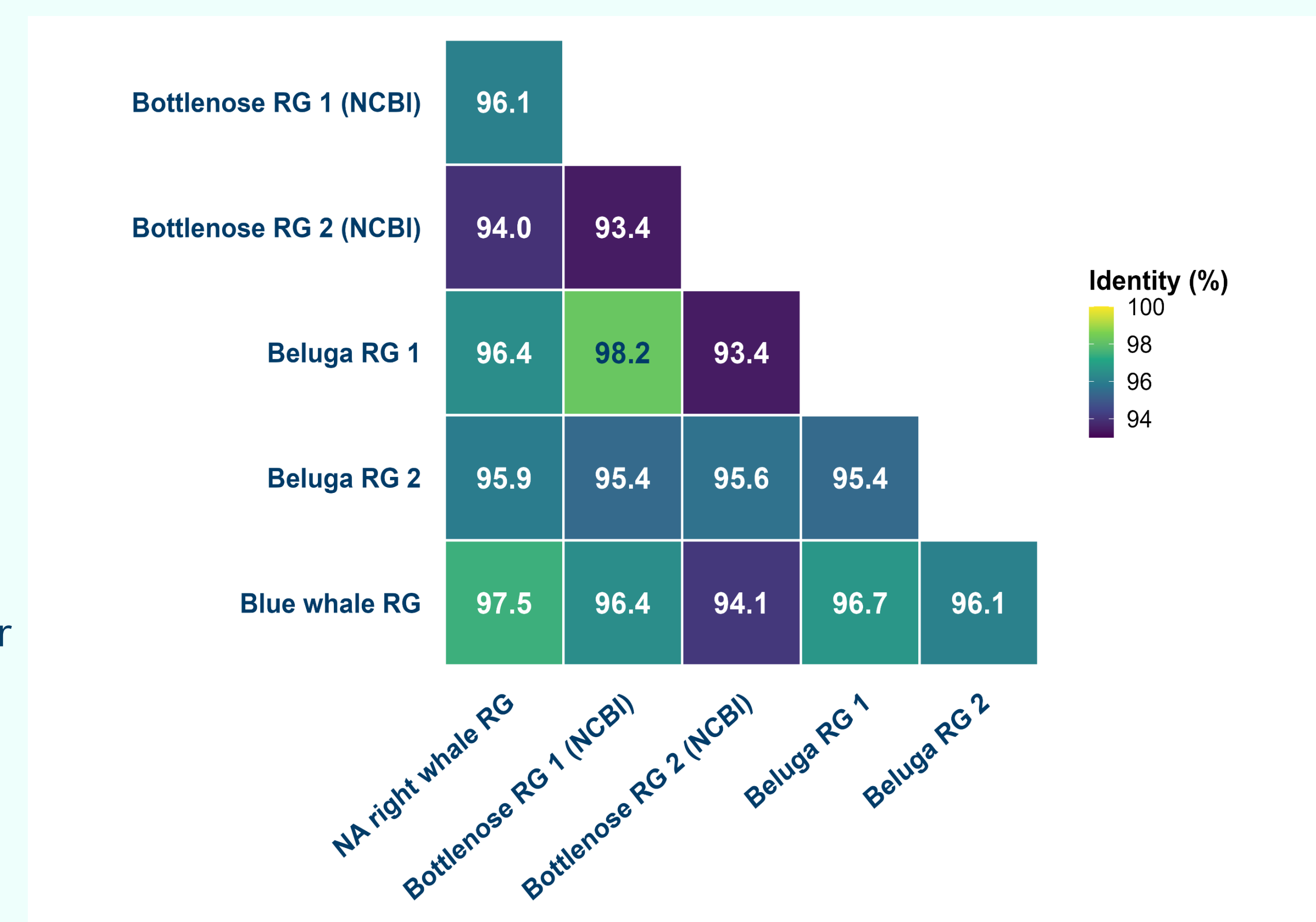


Figure: PCNA retrogene (RG) identity calculated at aligned positions where both sequences contained a nucleotide. Each unique sequence pair is shown once, and self-comparisons were omitted for clarity

## DISCUSSION & CONCLUSION

- A pattern seems evident in the number of copies in baleen vs. toothed cetaceans.
- Targeted approach of long-range PCR followed by Oxford Nanopore amplicon sequencing seems to result in confident identification of copies.

### Where do we go from here?

- More species! Of particular interest are the bowhead whale and the *Physeter macrocephalus* (sperm whale), that seem to be curious cases
- Functional assays using cell lines: *what exactly are these PCNA copies doing in the cell?*
- Analysis of predicted proteins: *which variations might alter function more?*

Patterns in retrotransposition can provide insight into previously established evolutionary timelines: whether copies evolved *before* divergence between toothed and baleen whales or *after*.

**Understanding of whether copy number variation contributed to Peto's paradox would broaden our understanding of cancer, which could ultimately make better therapeutics and treatments possible.**

## ACKNOWLEDGEMENTS

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## REFERENCES

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